Individual Tasks

My part which I individually contributed of the necessary tasks was to create a player movement controller for our project in Unity using C#. I first started by creating a PlayerController script that allows the player to move around the 3D LEGO world using the W, A, S and D keys.

```
using UnityEngine;
using UnityEngine.InputSystem;

[RequireComponent(typeof(CharacterController))]

public class PlayerController : MonoBehaviour
{
    [SerializeField] private float playerSpeed = 5f;
    private CharacterController controller;


    private void Awake()
    {
        controller = GetComponent<CharacterController>();
    }


    private void Update()
    {
        Keyboard keyboard = Keyboard.current;
        if (keyboard == null) return;

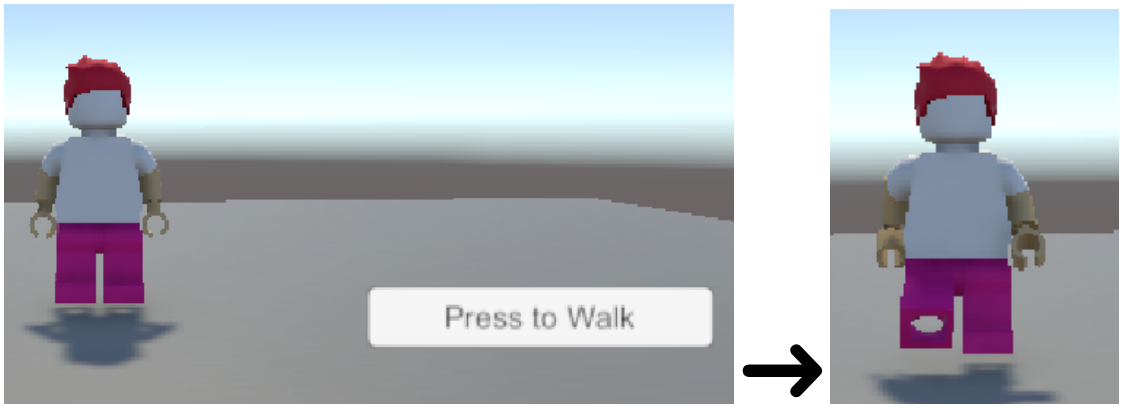
        float x = (keyboard.dKey.isPressed ? 1f : 0f)
                - (keyboard.aKey.isPressed ? 1f : 0f);

        float z = (keyboard.wKey.isPressed ? 1f : 0f)
                - (keyboard.sKey.isPressed ? 1f : 0f);

        Vector3 movement = transform.right * x + transform.forward * z;
        controller.Move(movement.normalized * playerSpeed * Time.deltaTime);
    }
}
```

I used Unity's CharacterController and Input System to manage movement and collisions. I also added an adjustable variable called playerSpeed and used Time.deltaTime to keep movement consistent across different frame rates. Doing this helped my members test their script such as mouseinput and physics as moving around helped them feel if their scripts were working as they should. This also improved my confidence with understanding player movement, especially as I had struggled with creating these types of scripts in previous projects.

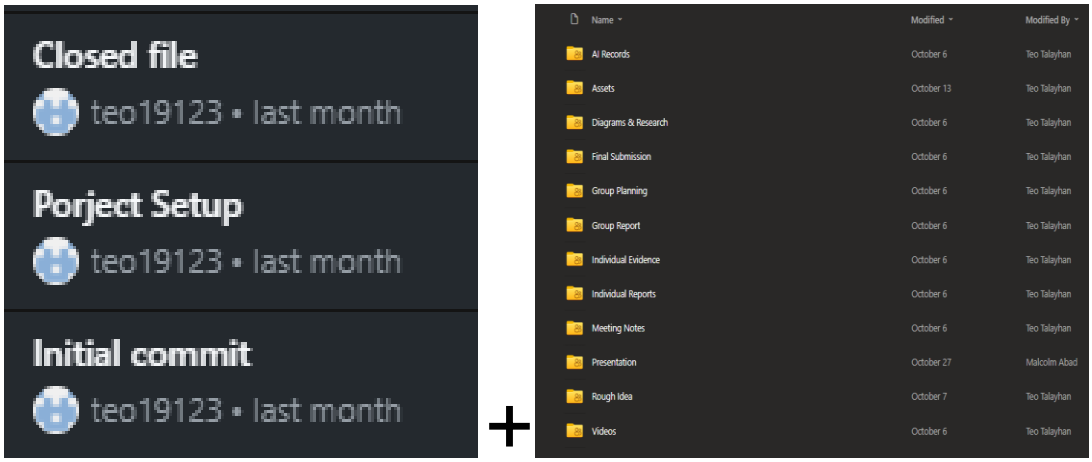
I also imported the LEGO minifigure to the Unity scene and used the Walk animation clip that was given to us for the task that was nesscary. I added an Animator to the character so Unity can play the animation. I added a button on the Canvas and connected its On Click event to the Animator's Play ("Walk") command. This is so, when the player presses that button, it will display the figure walking in place.



When first implementing it, a problem I faced was as soon as the animation started, the character turned away from the camera and changed its location. This was because the clip was a rotation and a root movement. I put the character under a parent anchor to hold it in place. Then I added a FaceCamera script to the animated character. This script causes it to change its path toward the Main Camera every frame even when the animation is playing. Finally, I checked the button in Play mode to make sure the Walk animation started and the character was still facing the camera and showing the intended animation.

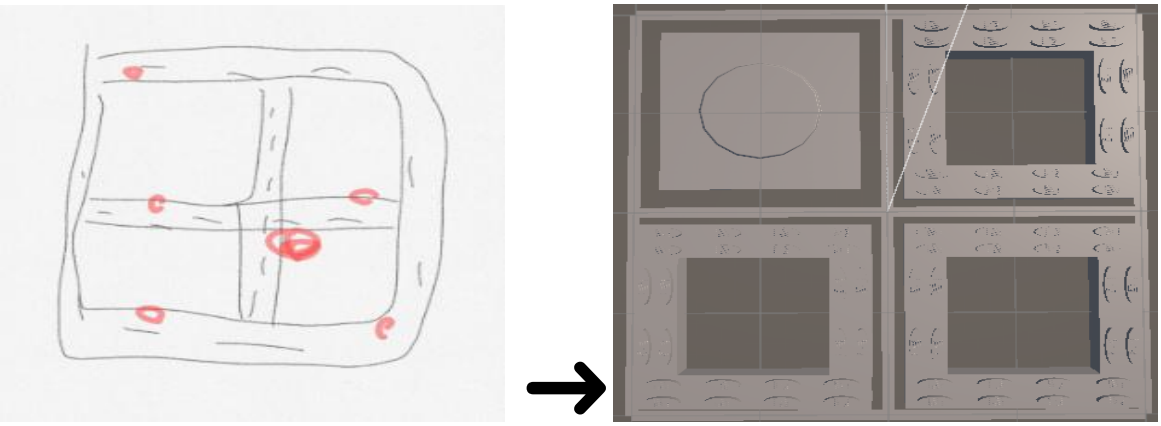
Group Tasks

My initial group task was to set up the shared files, assisted with project organization, and ensured that everyone had access to the materials they needed for participation. Creating the GitHub repository and sharing it with the team was my first task I completed. Additionally, I developed and established the primary Unity project, which evolved into a shared project to which everyone would then add their own work. This helped us make it simpler to integrate everyone's individual input, keeping all our work in one spot.



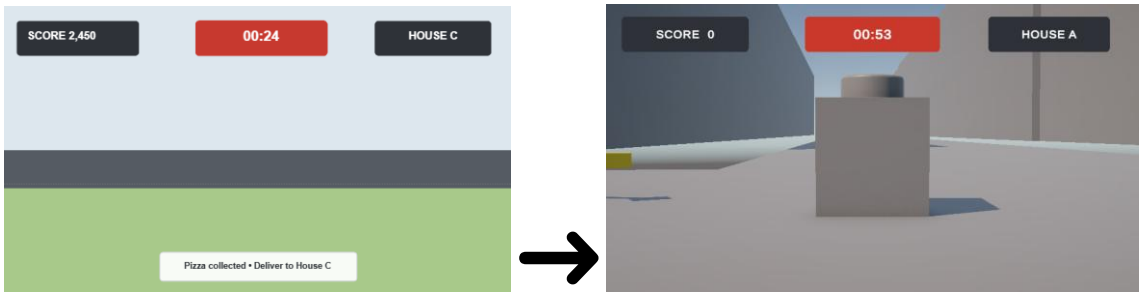
I was also an important contributor on the planning/documentation side of the project. I wrote the main game overview document, explaining the concept, gameplay idea, objectives, planned systems and further plans for the project. I produced all the written content, while some of the visual elements such as graphs and supporting images were created with help from Foyaz.

The basic level and map design was another thing I contributed to our project. I designed the environment's layout, including the location of the pizza shop, buildings, and roads. During a group discussion, I made a simple sketch, seen below which I then used as inspiration for our games layout by creating a crossroad separating the buildings, being the games simple road system.



The reason why we settled for this crossroad idea is since we don't want to overwhelm the player with a substantial number of controls and instructions. If we added a more complicated environment, this would have caused the player to neglect certain mechanics such as the HUD or score as they would be constantly getting lost in alley ways and junctions. Therefore, the primary goal at this point was to establish the foundation that could be further improved upon, the design was kept straightforward and simple.

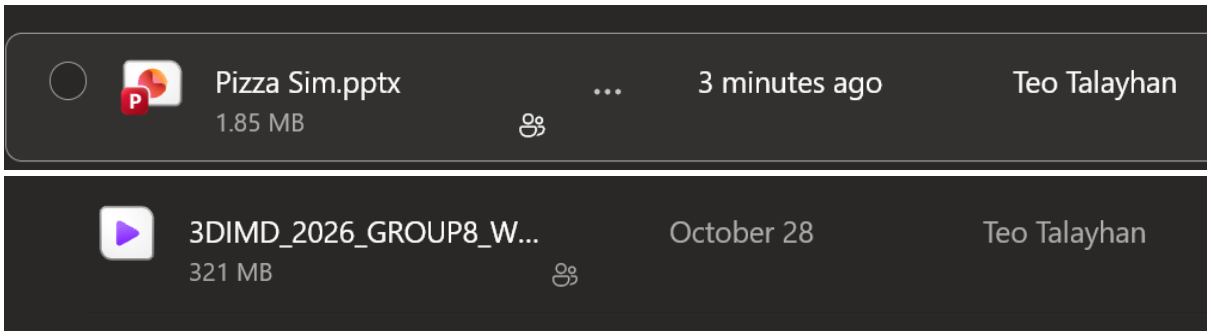
To demonstrate the player how information could potentially be shown, I also made a simple HUD design. Placeholders for components like the timer, score, and delivery details were added were first discussed then were built in. I had an advantage creating a HUD as I have had done this many times before in previous projects. One area which I did struggle with were creating the scripts for the timer, score and delivery checkpoints as these were new to me unlike the UI place holders.



I intended the HUD to be displays of the necessary information for the player so that they could use this to their advantage during gameplay to provide them ways to better succeed for faster deliveries and how they could be further expanded in the future.

For the group presentation I worked on the main presentation and the short video presentation explaining the project concept, current progress and what will be added. I also made sure to structure the information in a that captures our description of the project in an understandable way. I feel this helped us a lot during this phase as didn't

exactly know where to start, however after my contributions and illustrations, we had us our own framework to base the foundation of our project off.



For my asset contribution I designed my own LEGO character with Mecabricks. I wanted the character to blend in with the general theme of our pizza delivery simulator, so I went with the idea of a pizza chef. But I also wanted to incorporate a humorous touch to make the character unique and enjoyable. I just watched Pirates of the Caribbean recently and was influenced by the movie; I took the look of a pirate and blended it with a classic pizza chef. This made a funny pirate-chef character that fit the LEGO style of our game. I then modified the character for use in Unity, which gave me experience with external 3D modelling applications and asset integration.



I created this LEGO minifigure through the Mecabricks editor where you can create your own projects. I then went for legs which were formal but also fitted those of a chef. For the torso, I searched up chef which then display a body with a red handkerchief which is a sigh of those who work in cullinary. Then I added a pirate face to capture my inspiration and to finish off, I used the standard chef's hat.

One problem I faced was being unable to transfer my Mecabricks assets into unity for some reason. I overcame this issue by transferring it to Blender to the Mecabricks extension for Blender, and from there transferred it to the project.

As for AI use, I just used AI to help me better understand the CW1 coursework brief, especially the requirements and marking criteria. This enabled me to understand what

needed to be done to be able to get the most marks possible and to make sure that our group was working toward the right objectives.

Overall, I think my individual contributions spanned a lot of areas inside the project. I feel as I was working efficiently and well early in the project, however I felt overwhelmed at times as most of the workload fell onto my hands as some members didn't do their work tasks due to issues outside of school.